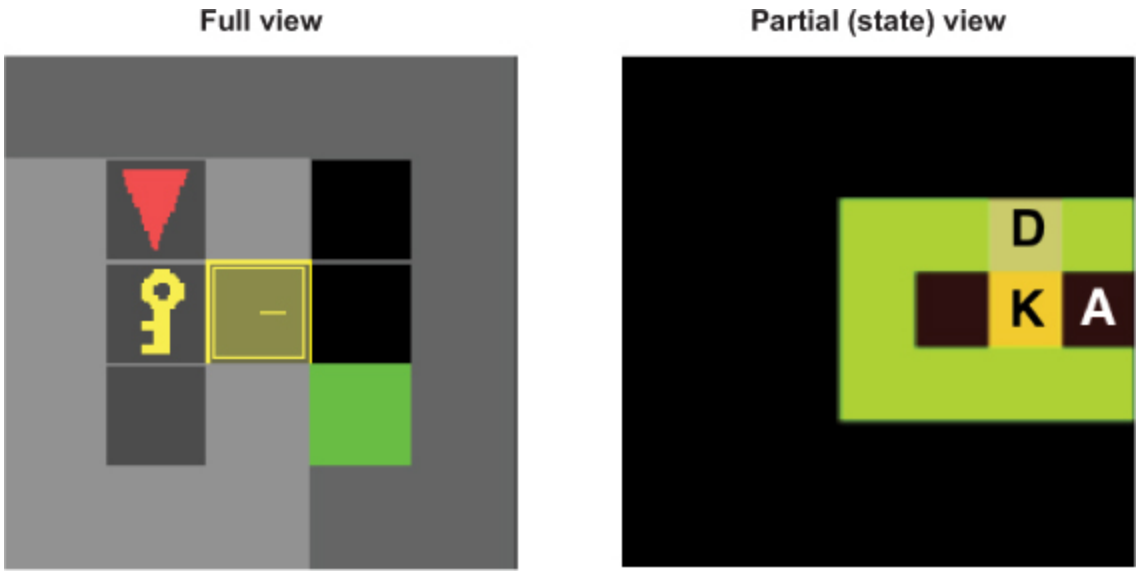


The partial view is a little confusing at first because it is an egocentric view, so we annotated it with the positions of the agent (A), key (K), and the door (D). Because the agent's partial view is always 7 × 7 and the size of the full grid is only 5 × 5, the partial view always includes some empty space. Now let's visualize the corresponding attention weights for this state.

In figure 10.24, each column is labeled as a particular node's attention weights (i.e., the nodes to which it is paying attention), restricting ourselves to just the agent, key, and door nodes out of a total of 7 × 7 = 49 nodes. Each row is an attention head, from head 1 to head 3, top to bottom. Curiously, attention head 1 does not appear to be focusing on anything obviously interesting; in fact, it is focusing on grid cells in empty space. Note that while we're only looking at 3 of the 49 nodes, even after looking at all of the nodes, the attention weights are quite sparse; only a few grid cells at most are assigned any significant attention. But perhaps this isn't surprising, as the attention heads appear to be specializing.

Figure 10.24. Each row corresponds to an attention head (e.g., row 1 corresponds to attention head 1). Left column: The self-attention weights for the agent, which shows the objects to which the agent is paying most attention. Middle column: The self-attention weights for the key, which shows the objects to which the key is paying most attention. Right column: The self-attention weights for the door.